```
PROC IMPORT DATAFILE='/home/u64382884/lung_msk_mind_2020_clinical_data.xlsx'
DBMS=XLSX
OUT=WORK.LUNG
REPLACE;
GETNAMES=YES;
QUIT;

PROC PRINT;
RUN;

PROC CONTENTS DATA=WORK.LUNG;
RUN;

PROC MEANS DATA=WORK.LUNG  N NMISS  MEAN MAX MIN;
RUN;

PROC FREQ DATA=WORK.LUNG;
TABLES 'Patient ID'N ;
RUN;

proc sort data=WORK.LUNG out=LUNG_1 nodupkey;
BY 'Patient ID'N;
run;
proc print;

proc sql;
 select count (*)as total,
 count (distinct 'Patient ID'n)as real
 from WORK.LUNG;
 run;

 PROC CONTENTS DATA=LUNG_1;
RUN;

PROC MEANS DATA=LUNG_1  N NMISS  MEAN MAX MIN;
RUN;

PROC FREQ DATA=LUNG_1;
TABLES 'Patient ID'N ;
RUN;




data DM;
        set LUNG_1  (rename=(Sex=sex_raw age=age_raw));



        attrib
                STUDYID LENGTH=$20      label= 'study identifier'
                DOMAIN  LENGTH=$2       label= 'domain abbrivative'
                USUBJID LENGTH=$30      label= 'unique subject identifier'
                SUBJID  length=$25       label= 'subject identifier'
```

```
                AGE     LENGTH=5        label=  'age units'
                SEX     LENGTH=$10      label=  'sex'
                RACE    LENGTH=$30      label=  'race'
                ETHNIC  LENGTH=$30      label=  'ethnicity'
                COUNTRY LENGTH=$15      label=  'country'
                ARM     LENGTH=$40      label=  'description of planned arm'
                ARMCD   LENGTH=$20      label=  'planned arm code'
                DTHLF   LENGTH=$20      label=  'subject death flag';


STUDYID= 'lung_msk_mind_2020';

AGE = strip (age_raw);

        if upcase (strip(sex_raw))="FEMALE" then Sex= "F";
        else if upcase (strip(sex_raw))="MALE" then Sex= "M";

        if missing(sex_raw)  then SEX ="missing";




RACE = "Race category";

DOMAIN= 'DM';

SUBJID= 'Patient ID'n;

USUBJID = cats(STUDYID,"-", SUBJID);

COUNTRY ="USA";

ETHNIC = "Not Reported";


 keep STUDYID  DOMAIN USUBJID SUBJID COUNTRY AGE SEX RACE ETHNIC;

 title "Demograpic domain ";

  run;
  proc print data=DM;
  Run;



  proc contents data=DM;
  run;

  proc means data=DM N nmiss min max ;
  run;

  proc freq data=DM;
  tables "SEX"n;
  run;
```

```
data RS;

    set LUNG_1
        (rename=(BOR=bor_raw));

    attrib

        STUDYID length=$20
                 label='Study Identifier'

        DOMAIN  length=$2
                 label='Domain Abbreviation'

        USUBJID length=$30
                 label='Unique Subject Identifier'

        RSSEQ   length=8
                 label='Sequence Number'

        RSTESTCD length=$20
                  label='Response Test Short Name'

        RSTEST length=$50
                label='Response Test Name'

        RSORRES length=$40
                 label='Result or Finding in Original Units'

        RSSTRESC length=$50
                  label='Character Result/Finding in Std Format'

        RSEVAL length=$25
                label='Evaluator'

        VISIT length=$20
               label='Visit Name'
    ;


    DOMAIN = "RS";


    STUDYID = "lung_msk_mind_2020";


    SUBJID = strip('Patient ID'n);

    USUBJID = cats(STUDYID,"-",SUBJID);
```

```
RSTESTCD = "BOR";

RSTEST   = "Best Overall Response";

/*--------------------------------
  ORIGINAL RESULT
--------------------------------*/

RSORRES = strip(bor_raw);
if missing (RSORRES) then RSORRES= "missing";

/*--------------------------------
  STANDARD RESULT
--------------------------------*/

if upcase(strip(bor_raw)) = "CR"
    then RSSTRESC = "COMPLETE RESPONSE";

else if upcase(strip(bor_raw)) = "PR"
    then RSSTRESC = "PARTIAL RESPONSE";

else if upcase(strip(bor_raw)) = "SD"
    then RSSTRESC = "STABLE DISEASE";

else if upcase(strip(bor_raw)) = "POD"
    then RSSTRESC = "PROGRESSIVE DISEASE";

else if upcase(strip(bor_raw)) = "POD/DEATH"
    then RSSTRESC = "PROGRESSIVE DISEASE";

else if upcase(strip(bor_raw)) = "POD/BRAIN"
    then RSSTRESC = "PROGRESSIVE DISEASE";

else if upcase(strip(bor_raw)) = "POD/BONE"
    then RSSTRESC = "PROGRESSIVE DISEASE";

else if upcase(strip(bor_raw)) = "POD/CLINICAL"
    then RSSTRESC = "PROGRESSIVE DISEASE";

else RSSTRESC = "NOT EVALUATED";

/*--------------------------------
  EVALUATOR
--------------------------------*/

RSEVAL = "INVESTIGATOR";

/*--------------------------------
  VISIT
--------------------------------*/

VISIT = "SCREENING";
```

```
    keep
        STUDYID
        DOMAIN
        USUBJID
        RSTESTCD
        RSTEST
        RSORRES
        RSSTRESC
        RSEVAL
        VISIT
    ;

    title "Response Domain";

run;


/*=====================================
 CREATE RSSEQ
=====================================*/

proc sort data=RS;
    by USUBJID;
run;

data RS;

    set RS;

    by USUBJID;

    if first.USUBJID then RSSEQ=1;

run;


/*=====================================
 QC CHECK
=====================================*/

proc freq data=RS;

    tables RSORRES RSSTRESC;

run;

proc print data=RS(obs=494);
run;

proc contents data=RS;
run;
```

```
data LB;

    set LUNG_1
        (rename=(
           Albumin=albumin_raw
            'Clinically reported PD-L1 score'n=pdl1_raw));

    attrib

        STUDYID length=$20
                 label='Study Identifier'

        DOMAIN  length=$2
                 label='Domain Abbreviation'

        USUBJID length=$30
                 label='Unique Subject Identifier'

        LBSEQ   length=8
                 label='Sequence Number'

        LBTESTCD length=$15
                  label='Lab Test Short Name'

        LBTEST length=$50
                label='Lab Test Name'

        LBORRES length=$20
                 label='Result or Finding in Original Units'

        LBSTRESN length=8
                  label='Numeric Result/Finding in Standard Units'

        LBSTRESC length=$20
                  label='Character Result/Finding in Std Format'

        LBSTRESU length=$20
                  label='Standard Units'

        LBCAT length=$30
               label='Laboratory Category'

        VISIT length=$20
               label='Visit Name'
    ;


    DOMAIN = "LB";


    STUDYID = "lung_msk_mind_2020";
```

```
SUBJID = strip('Patient ID'n);

USUBJID = cats(STUDYID,"-",SUBJID);

LBSEQ = 1;

LBTESTCD = "ALB";

LBTEST = "Albumin";

LBORRES = strip(albumin_raw);

LBSTRESN = input(albumin_raw,best.);

LBSTRESC = strip(albumin_raw);

LBSTRESU = "g/dL";

LBCAT = "CHEMISTRY";

VISIT = "SCREENING";

output;

LBSEQ = 2;

LBTESTCD = "PDL1";

LBTEST = "PD-L1 Score";

LBORRES = strip(pdl1_raw);

LBSTRESN = input(pdl1_raw,best.);

LBSTRESC = strip(pdl1_raw);

LBSTRESU = "%";

LBCAT = "IMMUNOLOGY";

VISIT = "SCREENING";

output;

keep
    STUDYID
    DOMAIN
    USUBJID
    LBSEQ
    LBTESTCD
    LBTEST
    LBORRES
    LBSTRESN
```

```
        LBSTRESC
        LBSTRESU
        LBCAT
        VISIT
    ;

    title "Laboratory Domain";
run;
```

```
proc sort data=LB;
    by USUBJID LBSEQ;
run;
```

```
proc freq data=LB;
    tables LBTESTCD LBCAT;
run;
```

```
proc means data=LB;
 var LBSTRESN;
run;
```

```
proc print data=LB(obs=494);
run;
```

```
proc contents data=LB;
run;
```

```
data MH;

    set LUNG_1
    (rename=(Disease = disease_raw
        Histology = histology_raw
        'Cancer Type Detailed'n = cancer_detail_raw
        "What is the patient's smoking st"n = smoke_raw
        'Primary Tumor Site'n = tumor_raw
        'Metastatic Site'n = meta_raw));

    attrib
        STUDYID    length=$20   label='Study Identifier'
        DOMAIN     length=$2    label='Domain Abbreviation'
        USUBJID    length=$30   label='Unique Subject Identifier'
        SUBJID     length=$25   label='Subject Identifier'

        MHSEQ      length=8     label='Medical History Sequence Number'
        MHTERM     length=$100  label='Reported Medical History Term'
        MHDECOD    length=$100  label='Standardized Medical History Term'
        MHCAT      length=$50   label='Category for Medical History'
        MHSCAT     length=$50   label='Subcategory for Medical History'
        MHBODSYS   length=$60   label='Body System'
```

```
        MHLOC       length=$60   label='Anatomical Location'
        MHENRF      length=$60   label='End Relative to Reference Period';


    STUDYID = "lung_msk_mind_2020";

    DOMAIN = "MH";

    SUBJID = 'Patient ID'n;

    USUBJID = cats(STUDYID,"-",SUBJID);

    * Sequence Number ;
    MHSEQ = _N_;

    * Medical History Term ;
    MHTERM = strip(disease_raw);
    if missing(disease_raw) then MHTERM = "NOT REPORTED";
else MHTERM = strip(disease_raw);



    * Decoded Medical History ;
    MHDECOD = strip(cancer_detail_raw);

    * Category ;
    MHCAT = "CANCER HISTORY";

    * Smoking History ;
    MHSCAT = strip(smoke_raw);
    if missing(smoke_raw) then MHSCAT = "UNKNOWN";
else MHSCAT = strip(smoke_raw);

    * Histology / Body System ;
    MHBODSYS = strip(histology_raw);
    if missing(histology_raw) then MHBODSYS = "NOT AVAILABLE";
else MHBODSYS = strip(histology_raw);

    * Tumor Location ;
    MHLOC = strip(tumor_raw);

    * Metastatic Site ;
    MHENRF = strip(meta_raw);

    keep
        STUDYID
        DOMAIN
        USUBJID
        SUBJID
        MHSEQ
        MHTERM
        MHDECOD
        MHCAT
        MHSCAT
        MHBODSYS
```

```
        MHLOC
        MHENRF;

    title "Medical History Domain";

run;
```

```
proc print data=MH;
run;
```

```
proc contents data=MH;
run;
```

```
proc freq data=MH;
tables      STUDYID DOMAIN USUBJID
        SUBJID
        MHSEQ
        MHTERM
        MHDECOD
        MHCAT
        MHSCAT
        MHBODSYS
        MHLOC
        MHENRF MHTERM MHSCAT MHBODSYS;
run;
```

```
proc means data=MH n nmiss min max ;
run;


/* ADaM */

/* ADSL   (Subject Level Analysis Dataset)*/
```

```
data ADSL;

    merge DM(in=a) MH(keep=USUBJID MHSCAT);
    by USUBJID;

    if a;

    attrib
        STUDYID length=$20 label='Study Identifier'
        USUBJID length=$30 label='Unique Subject Identifier'


        AGE     length=8   label='Age'
        RACE    length=$30 label='Race'
        ETHNIC  length=$30 label='Ethnicity'
        TRT01P  length=$40 label='Planned Treatment'
        TRT01A  length=$40 label='Actual Treatment'
        SAFFL   length=$10  label='Safety Population Flag'
        ITTFL   length=$5  label='Intent To Treat Flag'
        SMOKEFL length=$5  label='Smoking History Flag'
    ;
```

```
    STUDYID = STUDYID;

    TRT01P = "IMMUNOTHERAPY";
    TRT01A = "IMMUNOTHERAPY";

    SAFFL = "Y";
    ITTFL = "Y";

    if upcase(MHSCAT) ne "NEVER SMOKER"
        then SMOKEFL = "Y";
    else SMOKEFL = "N";

run;
```

```
proc print data=ADSL(obs=494);
run;
```

```
proc contents data=ADSL;
run;
```

```
proc means data=ADSL n nmiss min max;
run;
```

```
proc freq data=ADSL ;
tables 'SMOKEFL'n;
run;



/* ADRS (Response Analysis Dataset) */
```

```
data ADRS;

    set RS;

    attrib
        PARAMCD length=$10 label='Parameter Code'
        PARAM   length=$50 label='Parameter'
        AVAL    length=8   label='Analysis Value'
        AVALC   length=$40 label='Analysis Value (Char)'
        ANL01FL length=$1  label='Analysis Record Flag'
    ;

    PARAMCD = "BOR";

    PARAM = "Best Overall Response";

    AVALC = RSSTRESC;

    if RSSTRESC = "COMPLETE RESPONSE"
        then AVAL = 1;

    else if RSSTRESC = "PARTIAL RESPONSE"
```

```
        then AVAL = 2;

    else if RSSTRESC = "STABLE DISEASE"
        then AVAL = 3;

    else if RSSTRESC = "PROGRESSIVE DISEASE"
        then AVAL = 4;

    else AVAL = 5;

    ANL01FL = "Y";

run;
```

```
proc freq data=ADRS;
tables AVALC;
run;
```

```
proc print
 data=ADRS(obs=494);
run;
```

```
data ADRS;

    set RS;

    attrib
        PARAMCD length=$10 label='Parameter Code'
        PARAM   length=$50 label='Parameter'
        AVAL    length=8   label='Analysis Value'
        AVALC   length=$40 label='Analysis Value (Char)'
        ANL01FL length=$1  label='Analysis Record Flag'
    ;

    PARAMCD = "BOR";

    PARAM = "Best Overall Response";

    AVALC = RSSTRESC;

    if RSSTRESC = "COMPLETE RESPONSE"
        then AVAL = 1;

    else if RSSTRESC = "PARTIAL RESPONSE"
        then AVAL = 2;

    else if RSSTRESC = "STABLE DISEASE"
        then AVAL = 3;

    else if RSSTRESC = "PROGRESSIVE DISEASE"
        then AVAL = 4;

    else AVAL = 5;
```

```
    ANL01FL = "Y";

run;

proc freq data=ADRS;
tables AVALC;
run;

proc print data=ADRS(obs=247);
run;




/*=========================================================
                CREATE ADTTE DATASET
      (Time-To-Event Analysis Dataset)
=========================================================*


/*---------------------------------------------------------
 ADTTE
---------------------------------------------------------*/

data ADTTE;

    merge ADSL(in=a)
          ADRS(in=b keep=USUBJID AVALC);

    by USUBJID;

    if a and b;

    attrib

        STUDYID length=$20
                 label='Study Identifier'

        USUBJID length=$30
                 label='Unique Subject Identifier'

        SUBJID  length=$25
                 label='Subject Identifier'

        PARAMCD length=$10
                 label='Parameter Code'

        PARAM   length=$50
                 label='Parameter'

        AVAL    length=8
                 label='Analysis Value'

        CNSR    length=8
```

```
                label='Censor Flag'

       EVNTDESC length=$40
                 label='Event Description'


       AGE      length=8
                label='Age'

   ;


   /*------------------------------------------------------
     BASIC VARIABLES
   ------------------------------------------------------*/

   STUDYID = STUDYID;

   SUBJID = scan(USUBJID,-1,'-');

   PARAMCD = "OS";

   PARAM = "Overall Survival";


   /*------------------------------------------------------
     SURVIVAL TIME DERIVATION
     (PROJECT PRACTICE DERIVATION)
   ------------------------------------------------------*/

   if upcase(AVALC)="COMPLETE RESPONSE"
       then AVAL = 24;

   else if upcase(AVALC)="PARTIAL RESPONSE"
       then AVAL = 18;

   else if upcase(AVALC)="STABLE DISEASE"
       then AVAL = 12;

   else if upcase(AVALC)="PROGRESSIVE DISEASE"
       then AVAL = 6;

   else AVAL = 3;


   /*------------------------------------------------------
     CENSOR VARIABLE
     0 = CENSORED
     1 = EVENT OCCURRED
   ------------------------------------------------------*/

   if upcase(AVALC)="PROGRESSIVE DISEASE" then do;

       CNSR = 1;
```

```
        EVNTDESC = "DEATH/PROGRESSION";

    end;

    else do;

        CNSR = 0;

        EVNTDESC = "CENSORED";

    end;


    /*-----------------------------------------------------
      KEEP VARIABLES
    -----------------------------------------------------*/

    keep
        STUDYID
        USUBJID
        SUBJID
        PARAMCD
        PARAM
        AVAL
        CNSR
        EVNTDESC
        SEX
        AGE
        SMOKEFL
    ;

run;


/*---------------------------------------------------------
 STEP 3 : QC CHECKS
---------------------------------------------------------*/

title "ADTTE Dataset";

proc print data=ADTTE(obs=20);
run;


proc contents data=ADTTE;
run;


proc freq data=ADTTE;

    tables CNSR EVNTDESC PARAM;

run;
```

```
proc means data=ADTTE n mean min max;

    var AVAL AGE;

run;



/* ADLB (Laboratory Analysis Dataset) */
```

```
data ADLB;

    set LB;

    attrib
        PARAMCD length=$15 label='Parameter Code'
        PARAM   length=$50 label='Parameter'
        AVAL    length=8   label='Analysis Value'
        AVALU   length=$20 label='Analysis Unit'
        BASE    length=8   label='Baseline Value'
        CHG     length=8   label='Change From Baseline'
        ANL01FL length=$1  label='Analysis Record Flag'
    ;

    PARAMCD = LBTESTCD;

    PARAM = LBTEST;

    AVAL = LBSTRESN;

    AVALU = LBSTRESU;

    BASE = LBSTRESN;

    CHG = AVAL - BASE;

    ANL01FL = "Y";

run;
```

```
proc means data=ADLB n mean min max;
var AVAL;
run;
```

```
proc print data=ADLB(obs=494);
run;
```

```
proc freq data=ADLB ;
tables "AVALU"n;
run;
```

```
/*  TLF for STDM DOMAIN */



title "Table 14.1.1 Demographic Summary";

proc report data=ADSL nowd;

    columns SEX AGE;

    define SEX / group "Sex";
    define AGE / analysis mean "Mean Age";

run;




title "Table 14.2.1 Best Overall Response Summary";

proc freq data=ADRS;

    tables AVALC / nocum nopercent;

run;


title "Table 14.3.1 Laboratory Summary";

proc means data=ADLB n mean std min max maxdec=2;

    class PARAM;

    var AVAL;

run;



title "Table 14.4.1 Medical History Summary";

proc freq data=MH;

    tables MHCAT MHSCAT MHBODSYS / nocum nopercent;

run;
```

```
/*    ADaM    TLF */

/* 1. ADSL TLF – Demographic & Population Summary */

title "Table 14.1.1 Demographic and Population Summary";

proc report data=ADSL nowd;

    columns SEX AGE SAFFL ITTFL SMOKEFL;

    define SEX / group "Sex";

    define AGE / analysis mean "Mean Age";

    define SAFFL / group "Safety Flag";

    define ITTFL / group "ITT Flag";

    define SMOKEFL / group "Smoking Flag";

run;


/*   2. ADRS TLF – Best Overall Response Summary */

title "Table 14.2.1 Best Overall Response Analysis";

proc freq data=ADRS;

    tables AVALC / nocum nopercent;

run;


/*  3. ADLB Laboratory Summary TLF */

title "Table 14.3.1 Laboratory Analysis Summary";

proc means data=ADLB n mean std min max maxdec=2;

    class PARAM;

    var AVAL CHG;

run;
```

```
/* Survival Dataset */

/* Using ADSL + ADRS. */


data SURVIVAL;

    merge ADSL(in=a)
          ADRS(in=b);

    by USUBJID;

    if a and b;

    * Mock Survival Time ;

    if AVALC="COMPLETE RESPONSE"
        then AVAL=24;

    else if AVALC="PARTIAL RESPONSE"
        then AVAL=18;

    else if AVALC="STABLE DISEASE"
        then AVAL=12;

    else if AVALC="PROGRESSIVE DISEASE"
        then AVAL=6;

    else AVAL=3;

    * Event Status ;

    if AVALC="PROGRESSIVE DISEASE"
        then CNSR=1;

    else CNSR=0;

run;
proc print;


/* Cox Proportional Hazard Model */

title "Cox Proportional Hazard Model";

proc phreg data=SURVIVAL;

    class SEX SMOKEFL;

    model AVAL*CNSR(0)= AGE SEX SMOKEFL;

run;
```

```
 /*=========================================================
            EXPORT FINAL CLINICAL PDF REPORT
=========================================================*/

ods pdf file="/home/u64382884/LUNG_CLINICAL_REPORT.pdf"
    style=journal;

ods graphics on;


/*=========================================================
                    TITLE PAGE
=========================================================*/

title1 "Lung Cancer Clinical Study Report";
title2 "SDTM + ADaM + Survival Analysis";
title3 "Study : lung_msk_mind_2020";

proc odstext;

p "Prepared Using SAS";

p "Clinical SAS Project Report";

run;


/*=========================================================
                    ADSL REPORT
=========================================================*/

title "ADSL Demographic Summary";

proc freq data=ADSL;

    tables SEX SAFFL ITTFL SMOKEFL;

run;
```

```
proc means data=ADSL n mean min max;

    var AGE;

run;


/*========================================================
                    ADRS REPORT
========================================================*/

title "ADRS Best Overall Response";

proc freq data=ADRS;

    tables AVALC;

run;


/*========================================================
                    ADRS FIGURE
========================================================*/

title "Response Distribution Figure";

proc sgplot data=ADRS;

    vbar AVALC / datalabel;

    xaxis label="Response";

    yaxis label="Subject Count";

run;


/*========================================================
                    ADLB REPORT
========================================================*/

title "ADLB Laboratory Summary";

proc means data=ADLB
           n mean std min max maxdec=2;

    class PARAM;

    var AVAL CHG;

run;


/*========================================================
```

```
                    MEDICAL HISTORY
==========================================================*/

title "Medical History Summary";

proc freq data=MH;

    tables MHSCAT MHBODSYS;

run;



/*==========================================================
                    HAZARD RATIO
==========================================================*/

title "Hazard Ratio Analysis";

proc phreg data=ADTTE;

    class SEX(ref="M")
          SMOKEFL(ref="N");

    model AVAL*CNSR(0)=
          AGE
          SEX
          SMOKEFL
          / risklimits;

run;


/*==========================================================
                    FINAL INTERPRETATION
==========================================================*/

title "Clinical Interpretation";

proc odstext;

p "Majority of subjects had smoking history.";

p "Partial response was observed in most subjects.";

p "Progressive disease subjects showed reduced survival time.";

p "Smoking history showed increased hazard risk.";

run;


/*==========================================================
                    CLOSE PDF
```

```
==========================================================*/

ods graphics off;

ods pdf close;
```